

Chapter 156 — Pathology of Prostatic Neoplasia — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- Isolated high-grade prostatic intraepithelial neoplasia (HGPIN), even multifocal → **no repeat biopsy**.
- Intraductal carcinoma (IDC-P) + grade group (GG) 1 → **not for active surveillance (AS)**; >80% upgraded.
- Atypical, suspicious → ~40% cancer; PSA does not stratify.
- Atypical → **expert review before re-biopsy**.
- T1a ≤5% vs T1b >5% of TURP (transurethral resection) chips.
- GG: 1 = 3+3; 2 = 3+4; 3 = 4+3; 4 = 8; 5 = 9-10.
- Pure 3+3 → **no nodal metastatic potential**.
- Report % pattern 4 for GG 2-3 on biopsy.
- Focal vs nonfocal positive margin: **3 mm**.
- Hormone-treated cancer → **no Gleason score**; cryo/HIFU residual → graded.

PROSTATIC INTRAEPITHELIAL NEOPLASIA

Cut-offs:

- HGPIN on biopsy ~4-5% (reported 0-25%).
- TMPRSS2:ERG: ~20% HGPIN vs ~50% cancers.
- IDC-P alone on biopsy <0.06%.
- IDC-P prevalence: ~2% low-risk → ~56% metastatic; ~60% post-ADT (androgen deprivation therapy)/chemotherapy.

Gold standard / rules:

- Low-grade PIN (LGPIN) **never reported**.
- HGPIN on TUR → **no further sampling**.
- PINATYP → manage as **atypical**.
- Atypical intraductal proliferation (AIP), benign or GG1 biopsy → **consider re-biopsy**.
- IDC-P alone → definitive therapy acceptable (or re-biopsy).
- Cribriform HGPIN no longer diagnosed → IDC-P or AIP.

Eponym:

- Guo-Epstein criteria — stringent, highly specific IDC-P definition.

ADENOCARCINOMA

Staging:

- T1c = non-palpable, needle biopsy-detected.
- pT2+ = positive margin at intraprostatic incision or ill-defined apex.
- pT3a = EPE (extraprostatic extension), incl. **microscopic bladder neck**.
- pT3b = seminal vesicle **muscular wall** invasion.

Location / spread:

- Peripheral zone posterior; **85% of T1c**; anterior ~15%.
- Anterior: same BCR (biochemical recurrence) risk; more positive margins with EPE.
- Multifocal >85%; ~20% contralateral adverse pathology.
- EPE via perineural spaces, posterior/posterolateral.
- SVI (seminal vesicle invasion) **commonest route**: out at base → periseminal tissue.
- Metastases: nodes, bone → lung (with bone) → liver, adrenal.
- Left supraclavicular node may be presenting site.

Volume:

- EPE uncommon <0.5 cm³; nodes/SVI uncommon <4 cm³.
- Transition zone tumours exit at larger volumes.

Grade:

- Biopsy score = **commonest + highest** (2005).
- Lowest in practice 3+3=6.
- GG4 includes 3+5 and 5+3.
- Grade groups: WHO 2016, CAP, AJCC endorsed.
- 5-yr BCR-free by GG 1-5: 96, 88, 63, 48, 26%.
- Nodes at RP (radical prostatectomy): 3+3 0%, 3+4 0.6%, 4+3 4.3%.

Upgrading:

- Biopsy GG1 → RP: ~1/3 upgraded; GG2 → GG3 ~20%.
- Very low-risk AS (1-2 cores, ≤50%/core, PSAD <0.15) → ~15%.
- GG1 + negative mpMRI → ~20% upgraded at 6 years.
- Predictors: more cancer, higher PSA, **smaller prostate**, fewer cores.

Needle biopsy:

- Separate/inked cores → score per core; pooled → one score.
- Max 2-3 cores per block.
- Atypical ~5% of biopsies; 80% of cancers found = GG1, 5% GG4-5.
- Report: **positive cores + one extent measure**.
- Perineural invasion on biopsy → **report**.
- GG3 60% pattern 4 → may skip neoadjuvant hormones; **90%** → give.

TUR:

- 6-8 cassettes → all T1b; 8-10 → **>90% T1a**.
- Embed all if <65 years (some centres).
- Adenosis: transition zone mimic; **no cancer risk**.

Radical prostatectomy:

- Grade each nodule; dominant nodule usually worst.
- Tertiary pattern 5 <5% → stays in its GG.
- SVI → **65% 5-yr progression**.
- Positive margins ~15%; only ~50% progress.
- Intraprostatic incision = risk of focal EPE + positive margin.
- Node-positive, 15-yr: BCR-free ~7%, metastasis-free ~42%, CSS ~58%.
- Independent predictors: grade, EPE extent, margins, SVI, nodes.
- Vascular invasion: 7% of <4 cm³; independently prognostic.

Treatment effect:

- Post-radiotherapy: benign / cancer no effect (graded) / cancer with effect (not graded = no cancer).
- Radiation atypia persists up to **72 months**.
- Finasteride → no consistent histological change.

SUBTYPES OF PROSTATE ADENOCARCINOMA

- Mucinous = acinar behaviour.
- Small cell: survival <1 yr; ~50% mixed; **not graded**.
- Ductal: pure 0.4-0.8%, mixed ~5%; **pattern 4** (PIN-like 3; comedonecrosis 5).
- Ductal: verumontanum papillary mass, **normal PSA/DRE**; visceral recurrence.
- Squamous: osteolytic, hormone-unresponsive.
- Sarcomatoid: dismal.

Urothelial carcinoma:

- 35-45% of cystoprostatectomies involve prostate.
- Intraductal only → no prognostic effect.
- From prostatic urethral CIS: suburethral pT1; stroma pT2.

- Direct from bladder → pT4.
- GATA3 best urothelial marker; prostate: PSA, P501S, NKX3.1.

Mesenchymal / other:

- Sarcomas 0.1–0.2%.
- Commonest: child rhabdomyosarcoma; adult leiomyosarcoma.
- Inflammatory myofibroblastic tumour mimics leiomyosarcoma; post-TUR.
- STUMP → follow-up; resect if young.
- Leukaemia: commonest CLL/SLL.

CLASSIC TRAPS

- Trap: multifocal HGPIN → re-biopsy — **Actually:** no.

- Trap: GG1 + IDC-P → AS — **Actually:** excluded.
- Trap: bladder neck invasion = pT4 — **Actually:** microscopic = pT3a.
- Trap: 3+5 = GG5 — **Actually:** GG4.
- Trap: 3+4 ≈ 4+3 — **Actually:** GG2 vs GG3.
- Trap: Gleason 6 = mid-grade — **Actually:** lowest assigned.
- Trap: perineural invasion reported on RP — **Actually:** biopsy only.
- Trap: tumour volume independent at RP — **Actually:** no.
- Trap: tumour-to-ink distance matters — **Actually:** no.
- Trap: urothelial stromal invasion always pT4 — **Actually:** pT2 via urethral CIS.